

1. MLOps Concept

MLOps (Machine Learning Operations) refers to the practices that combine machine learning, DevOps, and data engineering to automate and manage the lifecycle of ML models. It is significant because it ensures models can be reliably deployed, monitored, and maintained in production environments with scalability and reproducibility.

2. AWS App Runner

AWS App Runner is a fully managed service that simplifies deploying web applications and APIs. In MLOps, it allows developers to deploy ML microservices directly from source code or container images without managing infrastructure, reducing operational complexity.

3. Benefits of Docker

Docker containers provide portability and consistency across environments. They ensure that models run the same in development and production. Another benefit is scalability, as containers can be easily replicated and orchestrated.

4. ECR vs Elastic Beanstalk

AWS ECR is a container registry used to store Docker images, while Elastic Beanstalk is a platform-as-a-service for deploying applications. Use ECR when managing container images, and Elastic Beanstalk when you want a simplified deployment without managing infrastructure.

5. CI/CD in MLOps

CI/CD automates testing and deployment of ML models. It ensures faster updates and reduces errors. GitHub Actions helps by automating workflows such as training, testing, and deployment pipelines.

6. IAM Roles

IAM roles define permissions for AWS services. They are crucial for security. For example, a SageMaker role may allow access to S3 for training data while restricting other services.

7. CloudWatch Monitoring

AWS CloudWatch monitors logs, metrics, and performance. It can track model latency, error rates, and resource usage, enabling real-time alerts and performance optimization.

8. Makefile Purpose

A Makefile automates repetitive tasks. In ML microservices, it may include commands like 'make build' or 'make deploy' to streamline workflows.

9. Docker Authentication to ECR

Authentication involves logging in to AWS and using a command like 'aws ecr get-login-password'. This is necessary to securely push Docker images to ECR.

10. Security Best Practices

Use IAM roles with least privilege, encrypt data, enable logging, secure endpoints, and regularly update dependencies to ensure safe ML deployments.